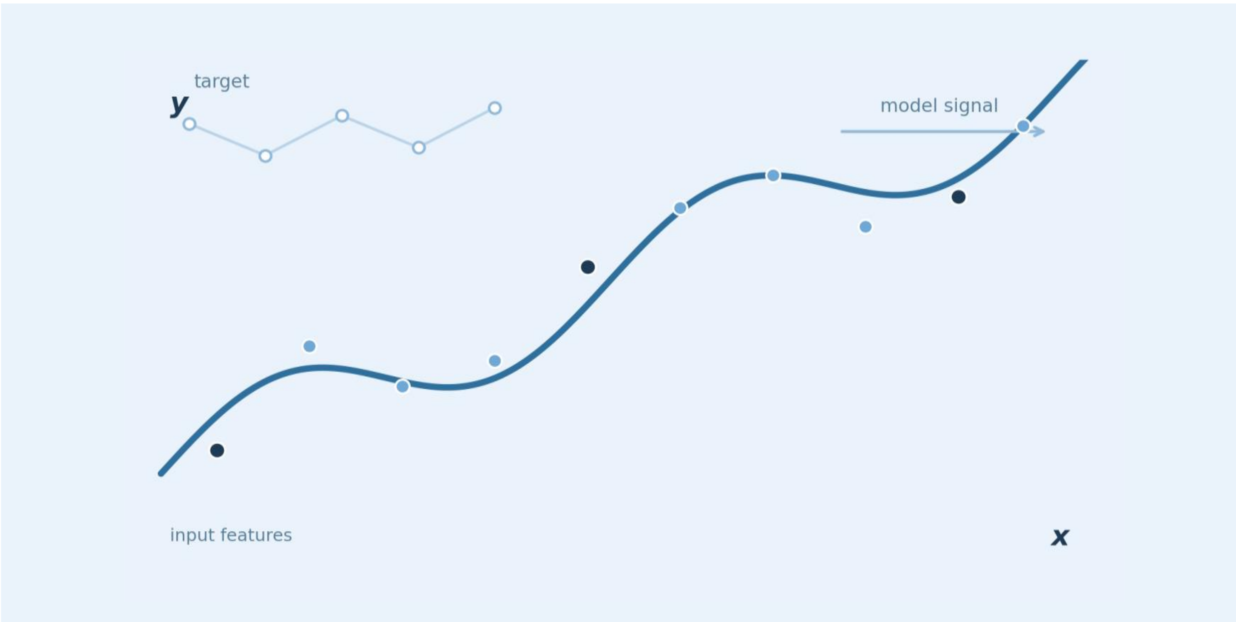
HOUSE PRICE PREDICTION

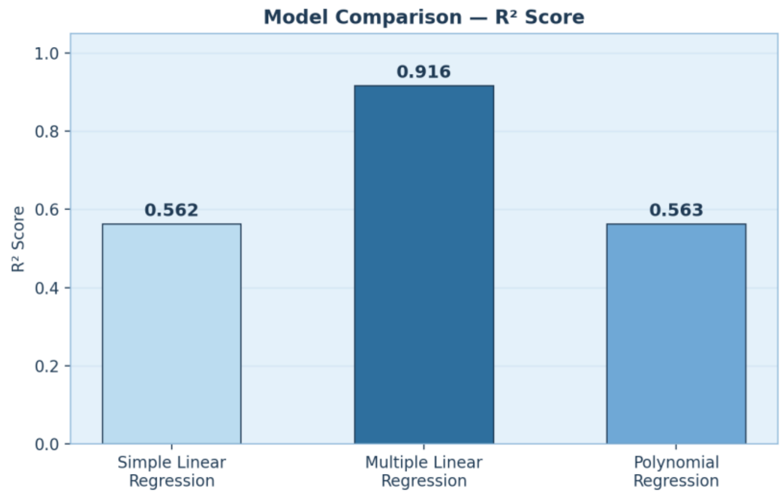
Final Analysis Report



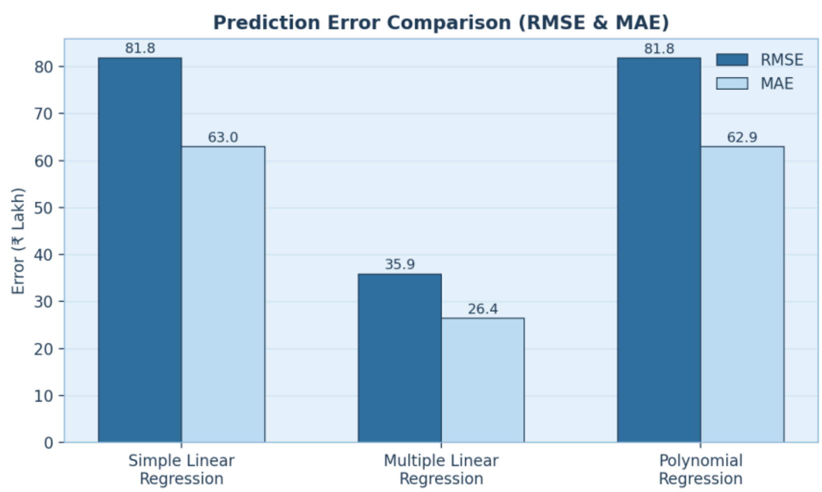
Machine Learning — Final Report

01 Best-Performing Model and Why

The **Multiple Linear Regression** model performed the best among the three models. It achieved an **R^2 score of 0.916**, while Simple Linear Regression and Polynomial Regression achieved around 0.563. Its RMSE and MAE were also much lower. MLR performed better because it uses more relevant features such as area, bedrooms, bathrooms, location score and age of the property, instead of depending mainly on area.



R^2 score comparison across the three regression models.



Multiple Linear Regression feature coefficients.

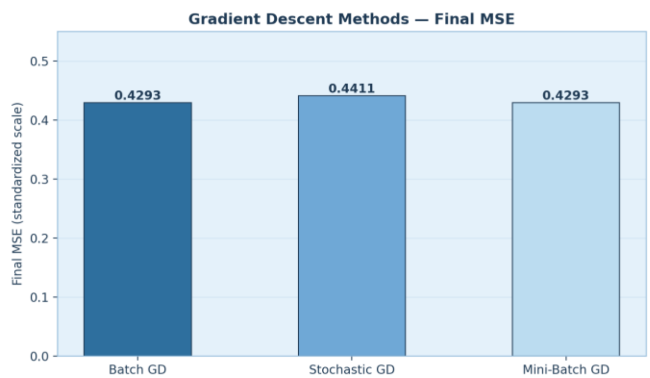
KEY IDEA MLR performed better because it uses more relevant features such as area, bedrooms, bathrooms, location score and age of the property, instead of depending mainly on area.

02 Impact of Gradient Descent Optimization

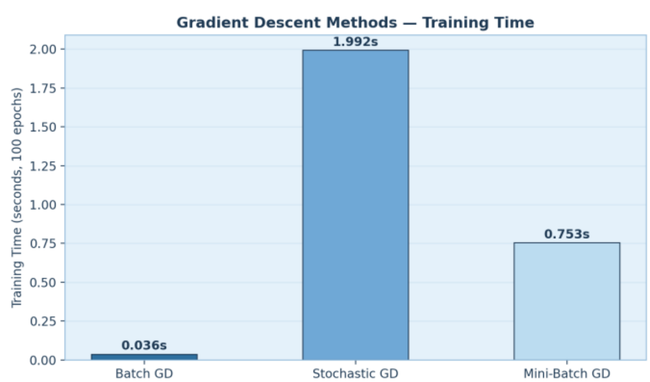
Gradient Descent was used to optimize the model parameters by reducing the error step by step. **Batch Gradient Descent** achieved a final MSE of 0.4293 with smooth convergence. **Mini-Batch Gradient Descent** achieved almost the same MSE, 0.4293, and gave a good balance between speed and stability. SGD also reduced the error but had a more noisy convergence pattern.

Method	Final MSE	Training Time (100 epochs)
Batch Gradient Descent	0.5073	0.036 s
Stochastic Gradient Descent	0.4314	1.992 s
Mini-Batch Gradient Descent	0.4293	0.753 s

Convergence and training-time comparison of the three gradient descent methods.



Final MSE achieved by each gradient descent method.

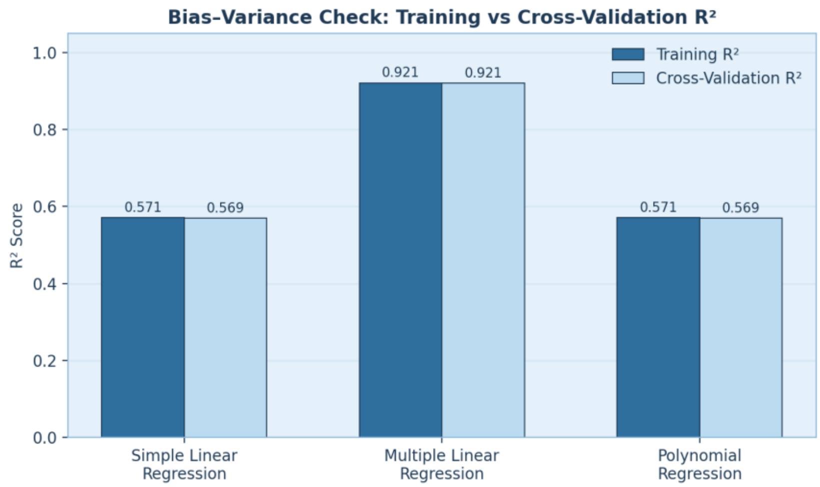


Training time comparison across gradient descent methods.

KEY IDEA Mini-Batch Gradient Descent achieved almost the same MSE, 0.4293, and gave a good balance between speed and stability.

03 Evidence of Overfitting / Underfitting

The results show that the Simple Linear Regression and Polynomial Regression models show some **underfitting**, because their R^2 scores are only around 0.57. The Polynomial Regression model did not show significant overfitting because its training and testing performance had a very small difference. The MLR model also had a very small **Train-CV gap (0.00047)**, showing good generalization.



Training R^2 vs. Cross-Validation R^2 for all three models.

Model	Training R^2	Mean CV R^2	Train–CV Gap
Simple Linear Regression	0.5707	0.5693	0.00134
Multiple Linear Regression	0.9211	0.9206	0.00047
Polynomial Regression	0.5708	0.5692	0.00160

Bias–variance diagnostics for all three models.

KEY IDEA The MLR model also had a very small Train-CV gap (0.00047), showing good generalization.

04 Practical Business Interpretation

This model can be useful for house price prediction. The results show that house price depends on more than just the area of the house. Factors such as **location, bedrooms, bathrooms and property age** provide additional information and improve prediction. Therefore, a real-estate business can use the model as a supporting tool for estimating house prices and comparing properties.

Input Features	→	MLR Model	→	Predicted House Price
area, bedrooms, bathrooms, location score, age		*learned relationship*		*supports business decisions*

KEY IDEA A real-estate business can use the model as a supporting tool for estimating house prices and comparing properties.

05 Final Conclusion

Overall, **Multiple Linear Regression** is the best model for this dataset because it provides the highest R^2 score and the lowest prediction errors among the tested models. The gradient descent experiments also showed that **Mini-Batch Gradient Descent** provides a good balance between accuracy, speed and stability.

REMEMBER Mini-Batch Gradient Descent provides a good balance between accuracy, speed and stability.